

CS215 Assignment-1

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1. Normal Sampling

Task A:

Solution: Consider the random variable $Y := F_X(X)$ in $[0, 1]$.

To find its distribution, we compute the CDF of Y , denoted by $F_Y(y)$. By definition:

$$F_Y(y) = P(Y \leq y) = P(F_X(X) \leq y).$$

ATQ, F_X is invertible, ie, there exists a function F_X^{-1} which is inverse of F_X .

Since $Y = F_X(X)$, the event $F_X(X) \leq y$ is equivalent to the event $X \leq F_X^{-1}(y)$. Therefore, we have:

$$F_Y(y) = P(X \leq F_X^{-1}(y)) = F_X(F_X^{-1}(y)) \quad (\text{by definition of CDF for } X)$$

$$\forall y \in [0, 1] \quad F_X(F_X^{-1}(y)) = y \implies F_Y(y) = y.$$

The CDF of Y is $F_Y(y) = y$ for $y \in [0, 1]$, which is exactly the CDF of a uniformly distributed random variable on $[0, 1]$.

Thus, Y is uniformly distributed on $[0, 1]$.

Task B:

Solution:

Assumption: We know the CDF of X .

Given a random variable Y uniformly distributed over $[0, 1]$, an algorithm $\mathcal{A}$ that transforms Y into a sample from a random variable with the CDF of X can be constructed as follows:

Let $Y \sim \text{Uniform}(0, 1)$. Suppose F_X is the CDF of the desired random variable X , which is continuous and invertible.

Algorithm $\mathcal{A}$:

Transform the uniform random variable Y by applying the inverse of the CDF of X .

$$\mathcal{A}(y) = F_X^{-1}(y).$$

Proof of Correctness:

Consider the CDF of $Z = F_X^{-1}(Y)$.

$$F_Z(u) = P(Z \leq u) = P(F_X^{-1}(Y) \leq u).$$

Since F_X is continuous and invertible, this is equivalent to:

$$P(F_X^{-1}(Y) \leq u) = P(Y \leq F_X(u)).$$

Because Y is uniformly distributed over $[0, 1]$, we have:

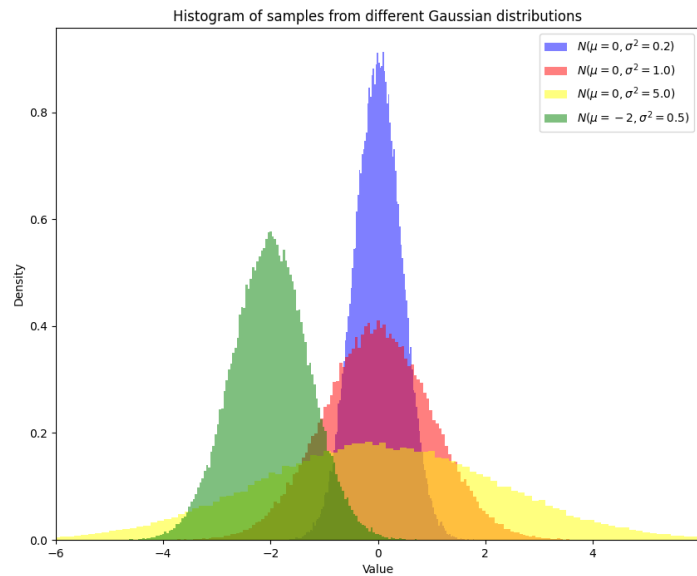
$$P(Y \leq F_X(u)) = F_X(u).$$

Thus, the CDF of Z is $F_Z(u) = F_X(u)$

Hence, Z (the output of the algorithm $\mathcal{A}$) has the same CDF as X .

Task C:

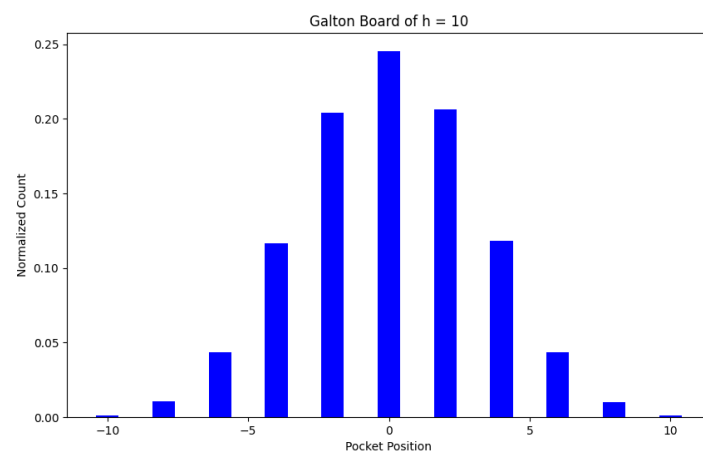
Solution:



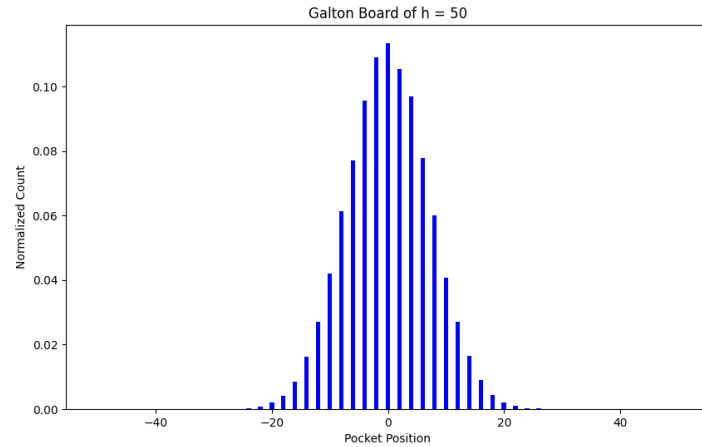
Bell Curves obtained using a Uniform RV

Task D:

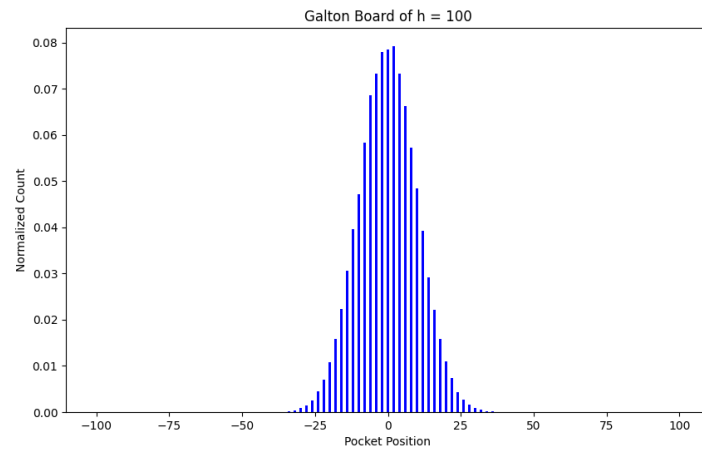
Solution: The histograms with the counts of balls in each pocket of the galton board for various values of h are as follows:



$h=10$



$h=50$



$h=100$

The shapes of these histograms closely approach the shape of a gaussian.

Task E:

Solution:

Each collision of the ball in a Galton Board has two possible outcomes: moving left or right. Hence, the motion of the ball can be modeled as a series of h independent Bernoulli trials with parameter $p = \frac{1}{2}$ (as both left and right are equally likely).

Now, a series of h Bernoulli trials can be represented as a binomial distribution with parameters h and $p = p_{\text{Bern}} = 1/2$. Let $R \sim \text{Bin}(h, 1/2)$ be the random variable representing the number of right moves out of h .

If the ball moves to the right r times out of h , the final position X can be expressed as:

$$X = r - (h - r) = 2r - h$$

Note that since h is even, X is even and can only take values in $\{-h, -h + 2, \dots, h - 2, h\}$.

The probability of landing in a specific pocket corresponding to position $X = 2i$ is given by:

$$P(X = 2i) = P(2R - h = 2i) = P(R = i + (h)/2)$$

Let $k = h/2$. From the binomial distribution,

$$\begin{aligned} P(X = 2i) &= \binom{h}{k+i} \left(\frac{1}{2}\right)^h \\ &= \left(\frac{1}{2}\right)^h \frac{h!}{(k+i)!(k-i)!} \end{aligned}$$

Now, Stirling's approximation for the factorial is $n! \approx \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$. Assuming that h is very large, and $i \ll h$, we can use Stirling's approximation for h , $k+i$, and $k-i$:

$$\begin{aligned} P(X = 2i) &\approx \left(\frac{1}{2}\right)^h \frac{\sqrt{2\pi h} \left(\frac{h}{e}\right)^h}{\left(\sqrt{2\pi(k+i)} \left(\frac{k+i}{e}\right)^{k+i}\right) \left(\sqrt{2\pi(k-i)} \left(\frac{k-i}{e}\right)^{k-i}\right)} \\ &= \sqrt{\frac{h}{2\pi(k^2 - i^2)}} \frac{h^h}{(h+2i)^{k+i} (h-2i)^{k-i}} \\ &\approx \sqrt{\frac{h}{2\pi k^2}} \frac{h^h}{((h+2i)(h-2i))^k} \left(\frac{h-2i}{h+2i}\right)^i \\ &= \sqrt{\frac{1}{\pi k}} \left(\frac{1}{1 - i^2/k^2}\right)^k \left(1 - \frac{4i}{h+2i}\right)^i \\ &\approx \sqrt{\frac{1}{\pi k}} \left(1 + \frac{i^2}{k}\right) \left(1 - \frac{2i^2}{k}\right) \\ &\approx \sqrt{\frac{1}{\pi k}} \left(1 - \frac{i^2}{k}\right) \\ &\approx \sqrt{\frac{1}{\pi k}} e^{-i^2/k} \end{aligned}$$

Hence, $P_h[X = 2i] \approx \frac{1}{\sqrt{\pi h}} e^{-\frac{i^2}{h}} = N(0, k/2)$.

This confirms that the distribution of the number of balls in each pocket approaches a normal distribution as h increases.